

Baligen Talihati

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🏠 balgen520.github.io ◇  [Google Scholar](#) ◇  [ResearchGate](#)

EDUCATION

University of Tennessee, Knoxville

Aug. 2025 – Present

Ph.D. Electrical Engineering and Computer Science

Advisor: [Hao Huang](#)

Research Area: Power Grid Resilience.

Fudan University

Sept. 2022 – Jun. 2025

M.S. Electronics Science and Technology, GPA 3.71/4.00

Core Modules: Modern Control Theory and Technology, Modern Sensor Technology, Detection of Weak Signal.

Thesis: Community Energy Management Method Based on Multi-Agent Reinforcement Learning.

Fudan University

Sept. 2016 – Jun. 2020

B.E. Electrical Engineering and Automation, GPA 3.16/4.00

Core Modules: Mathematical Analysis, C Programming Language, University Physics, Linear Algebra, Automatic Control Technology.

RESEARCH INTERESTS

Smart Grid, Renewable Energy, Machine Learning

PUBLICATIONS AND PATENTS

Publications:

[J.1] **Baligen Talihati**, Shengyu Tao, Shiyi Fu, Bowen Zhang, Hongtao Fan, Qifen Li, Xiaodong Lv, Yaojie Sun, Yu Wang, [Energy storage sharing in residential communities with controllable loads for enhanced operational efficiency and profitability](#), *Applied Energy* 373 (2024) 123880. doi:10.1016/j.apenergy.2024.123880. (JCR Q1 IF=11.0, Accepted on July 9, 2024)

[J.2] **Baligen Talihati**, Shiyi Fu, Bowen Zhang, Yuqing Zhao, Yu Wang, Yaojie Sun, [Community shared ES-PV system for managing electric vehicle loads via multi-agent reinforcement learning](#), *Applied Energy* 380 (2025) 125039. doi:10.1016/j.apenergy.2024.125039. (JCR Q1 IF=11.0, Accepted on November 25, 2024)

[J.3] Bowen Zhang, **Baligen Talihati**, Hongtao Fan, Yaojie Sun, Yu Wang, [A dynamic carbon flow traceability framework for integrated energy systems](#), *Journal of Cleaner Production* 518 (2025) 145878. doi:10.1016/j.jclepro.2025.145878. (JCR Q1 IF=10.0, Accepted on May 31, 2025)

Patents:

[P.1] Yu Wang, **Baligen Talihati**, Yaojie Sun, Bowen Zhang. Community energy storage sharing framework with power dispatch method. China National Intellectual Property Administration (CNIPA). Patent No.20230036. Filed September 6, 2023. Patent application accepted.

[P.2] Yaojie Sun, Yuanhui Chen, Yu Wang, **Baligen Talihati**, Bowen Zhang. Shared energy storage - multi-producer and consumer cooperative optimization method. China National Intellectual Property Administration (CNIPA). Patent No.20230003. Filed February 10, 2023. Patent application accepted.

RESEARCH EXPERIENCE

Community Energy Management Systems

Sept. 2023 – June. 2024

Graduate Researcher, Artificial Intelligence Energy Control Laboratory, Fudan University

Shanghai, CHN

- Prof. Yaojie Sun, Supervisor
- Developed a Community Energy Management System model using Python and Pytorch, based on Multi-Agent Reinforcement Learning.
- Evaluated effectiveness using metrics such as solar self-consumption and cost reduction.

- Enhanced solar self-consumption by 66.41% and reduced electricity costs by 7.73%, supporting a total load capacity of 38.68%.

Carbon Neutral Project in Industrial Park

Dec. 2022 – June. 2024

Graduate Researcher, State Grid Jiangsu Company

Nanjing, CHN

- Prof. Yaojie Sun, Supervisor
- Developed microgrid models under different cooperative storage modes.
- Assisted in a 100-million RMB project led by State Grid, ensuring smooth progress and timely delivery.
- Implemented shared storage-based systems at a factory site, assessing carbon footprint to support carbon neutrality.

WORK EXPERIENCE

Industrial AI Algorithm Application Intern, Siemens

Sept. 2024 – Mar.2025

Siemens Digital R&D Center

Shanghai, CHN

- Acquired proficiency in AI algorithms for industrial equipment anomaly detection, enhancing understanding of machine learning principles, types of industrial anomalies, and their root causes.
- Performed data processing tasks for equipment anomaly data, ensuring accurate analysis and timely insights for preventive measures.
- **Technical Proficiency:** Developed expertise in Python and scikit-learn libraries, utilizing advanced algorithms to improve system diagnostics and operational efficiency.

Substation Secondary Protection Maintenance, State Grid

July. 2020 – Oct. 2021

Urumqi Power Supply Company

Urumqi, CHN

- Responsible for the inspection and routine maintenance of secondary protection devices at 13 substations in Urumqi.
- Actively participated in daily company operations, including assisting with inspections and company promotion activities.
- **Technical Proficiency:** Developed a solid understanding of control circuits for secondary protection devices, cultivated strong communication and teamwork skills, and became proficient in PPT, video editing, and web presentation tools.

FELLOWSHIPS AND AWARDS

Shanghai Municipal Outstanding Graduate Award

Awarded for outstanding research performance

Sept. 2022 – Apr. 2025

Corporate-Sponsored Scholarship, approximately \$1100

Awarded for outstanding research performance

Sept. 2023 – Sept. 2024

Outstanding Student Scholarship, approximately \$1200

Awarded for outstanding academic performance

Sept. 2018 – Jun. 2019

TEACHING ASSISTANT

University of Tennessee, Knoxville

• ECE525 Alternative Energy Source

Fall 2025

• ECE581 High Frequency Power Electronics

Fall 2025

Fudan University

• INFO130263.01 Electronic Methods

Spring 2023

SKILLS & LANGUAGES

Proficient in Coding: Python, MATLAB, L^AT_EX.

Languages: Mandarin(Native), Kazakh Language(Native), English(IELTS: Overall 7.0)